MASTER OF BUSINESS ADMINISTRATION

SEMESTER 3

DADS304

VISUALIZATION

Unit 7

Editing, Building Views and Formatting

Table of Contents

SL No	Topic	Fig No / Table / Graph	SAQ / Activity	Page No
1	Introduction			3
	1.1 Learning Objective			
2	Edit Data sources and Extracts	1, 2, 3, 4, 5, 6, 7, 8	1	4-10
3	Replace and Metadata	9, 10, 11, 12, 13, 14, 15	2	11-15
4	Building Views and Formatting Views	16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29	3	16-26
5	Case studies			27-28
6	Terminal Questions	30, 31, 32, 33		29-31
7	Answers			32
8	Summary			32
9	Glossary			32
10	Concept map			32
11	Reference			33

1. INTRODUCTION

In Tableau Cloud or Tableau Server, you can make changes to a view if you can see the Edit button when viewing it. You can: Depending on your permissions and access level, edit a workbook that has already been published and include worksheets for views, dashboards, and stories.

A new worksheet should be created and edited using a published data source.

In the web or by launching the workbook in Tableau Desktop, you can edit an existing workbook and add worksheets. While editing, connect to several published data sources. See Connect to Published Data Sources while Web Editing for more information.

1.1 Learning Objective

- *Edit Data sources and Extracts*
- *Replace and Metadata*
- *Building Views and Formatting Views*

2. EDIT DATA SOURCES AND EXTRACTS

Tableau can connect to all of the most popular data sources. Tableau's native connectors can connect to the data sources listed below.

- File formats such as CSV, Excel, and others.
- Relational databases such as Oracle, SQL Server, DB2, and others.
- Cloud computing platforms such as Windows Azure, Google BigQuery, and others.
- Other ODBC-enabled sources

Live Connection

The Connect Live feature is used to analyse data in real time. Tableau connects to a real-time data source and continues to read the data in this case. As a result, the analysis result is up to the second, and the most recent changes are reflected in the result. However, the source system is burdened because it must continue to send data to Tableau.

In-Memory

Tableau can also process data in-memory by caching it in memory and not connecting to the source while analysing it. Of course, the amount of data cached will be limited by the amount of memory available.

Combine Data Sources

Tableau can connect to multiple data sources at once. For example, you can define multiple connections in a single workbook to connect to a flat file and a relational source. This is used in data blending, which is a very unique Tableau feature.

Tableau data extraction creates a subset of data from the data source. This is useful for improving performance by using filters. It also aids in applying Tableau features to data that may not be available in the data source, such as finding distinct values in the data. The data extract feature, on the other hand, is most commonly used to create an extract to be stored on the local drive for offline access by Tableau.

Steps to handle extracting data are as following:

1. Creating an Extract

Data extraction is accomplished by selecting Data Extract Data from the menu. It provides many options, such as limiting the number of rows extracted and deciding whether to aggregate data for dimensions.

2. Applying Extract Filters

Create filters that return only the relevant rows to extract a subset of data from the data source. Consider the Sample Superstore data set and extract it. Choose Select from list in the filter option and check the checkbox value for which you want to pull data from the source.

3. 3.Adding New Data to the Extract

To add more data to an already created extract, use the Data Extract Append Data from File option. In this case, browse to the data file and click OK to finish. Of course, the number and type of columns in the file must match the existing data.

4. Extract History: You can check the history of data extracts to see how many times the extract has occurred and when.

You can do this by going to the Data Extract History menu.

Practical steps:

You have the option to alter the workbook's data source at any time while conducting your study.

How to edit the data source

1. Select Edit Data Source after choosing a data source from the Data menu.
2. Make the necessary modifications to the data source on the data source page.

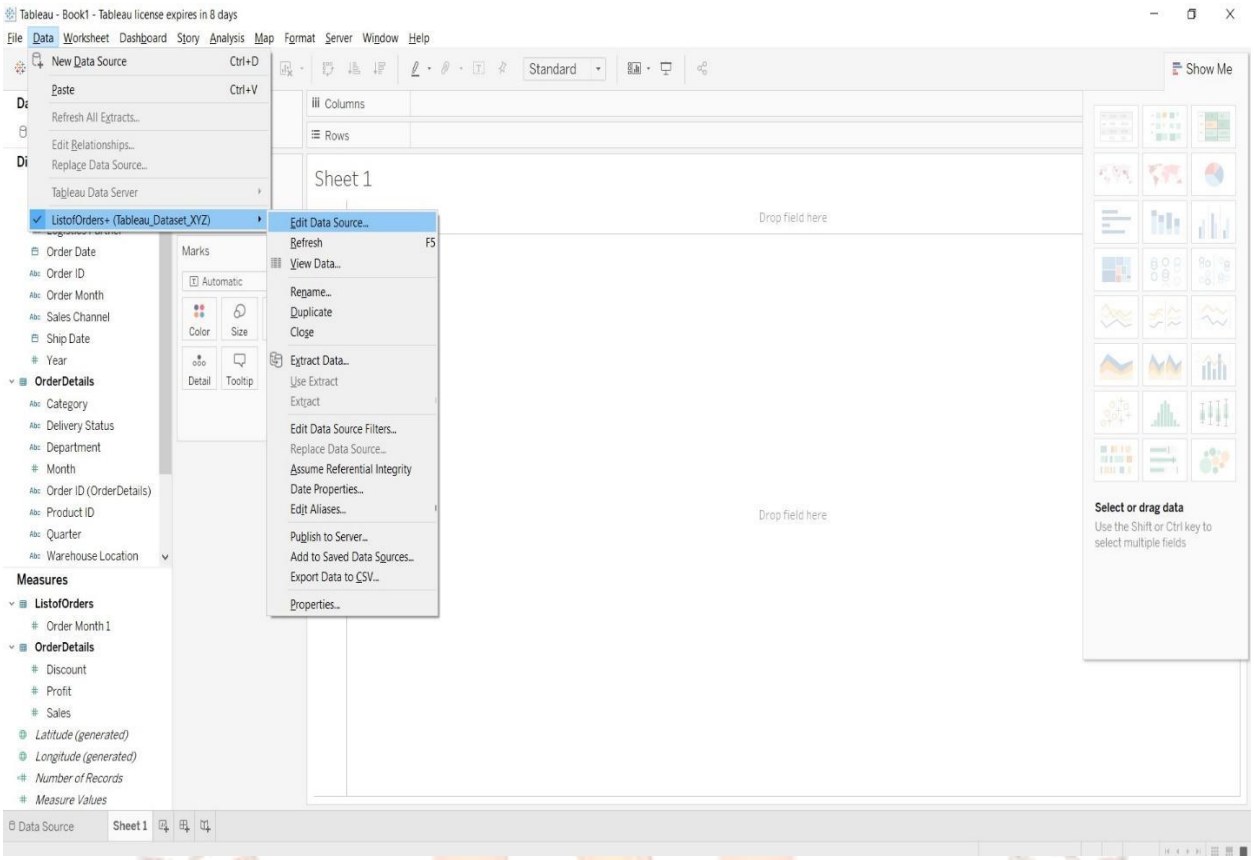


Fig 7.1: Edit Data Source

Perhaps you should change the data source to:

1. Gather Your Data
2. Link to a Special SQL Query (Tableau Desktop)
3. Employ a stored procedure (Tableau Desktop)

How to Extract Data

1. Go to the sheet tab,
2. Click on Data
3. Select Extract Data

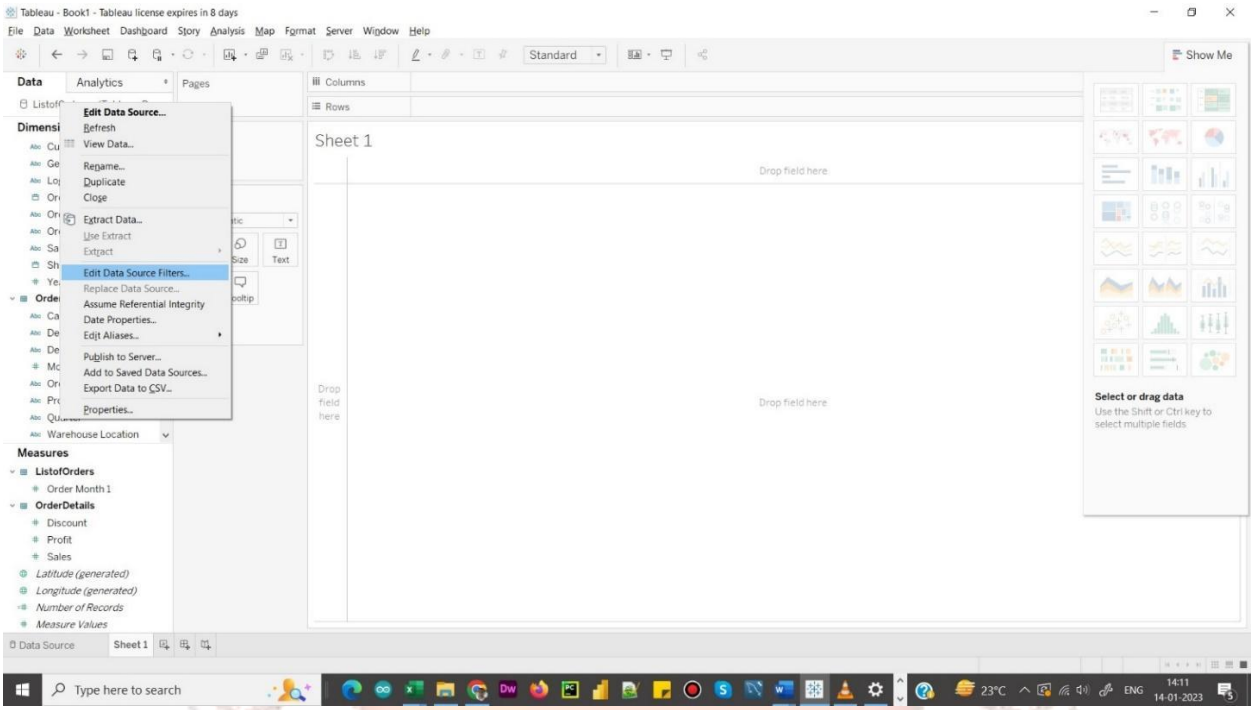


Fig 7.2: Edit Data Source from data

Here, you can extract data from a single table or many tables. You can also extract data based on the number of rows.

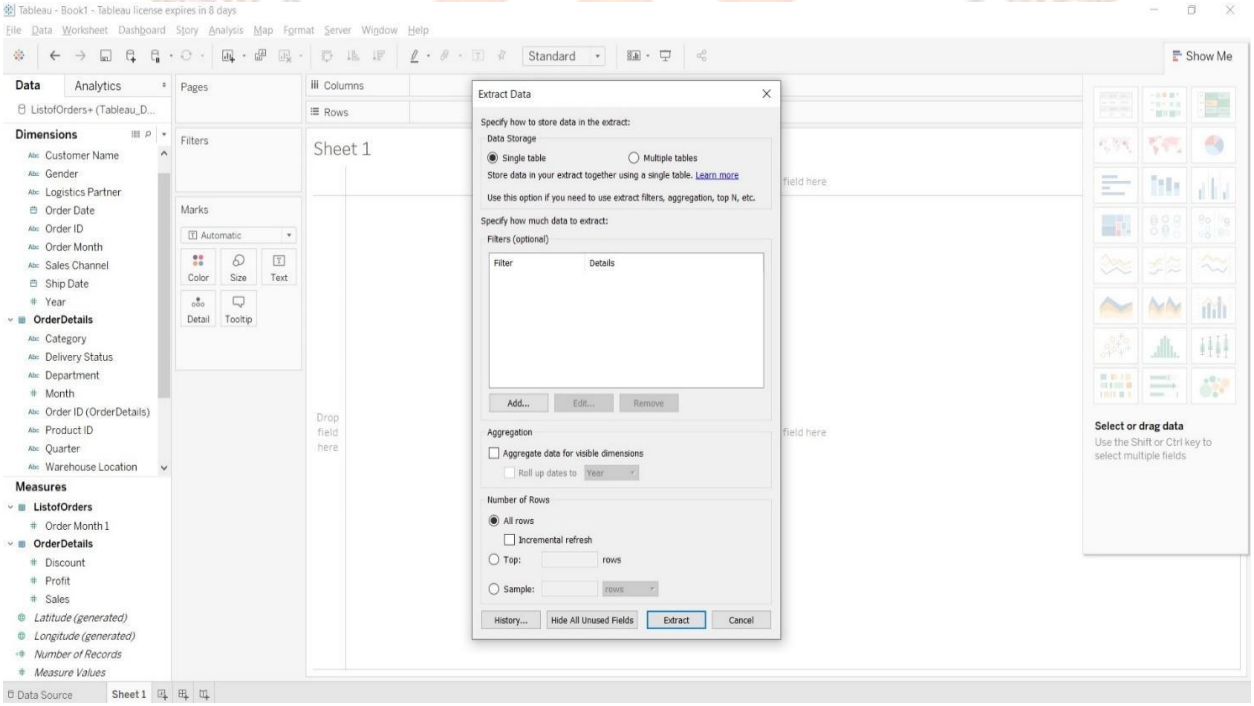


Fig 7.3: Extract specific data from dataset

There is an option to pick the number of rows from the dataset you want to extract. check the below diagram.

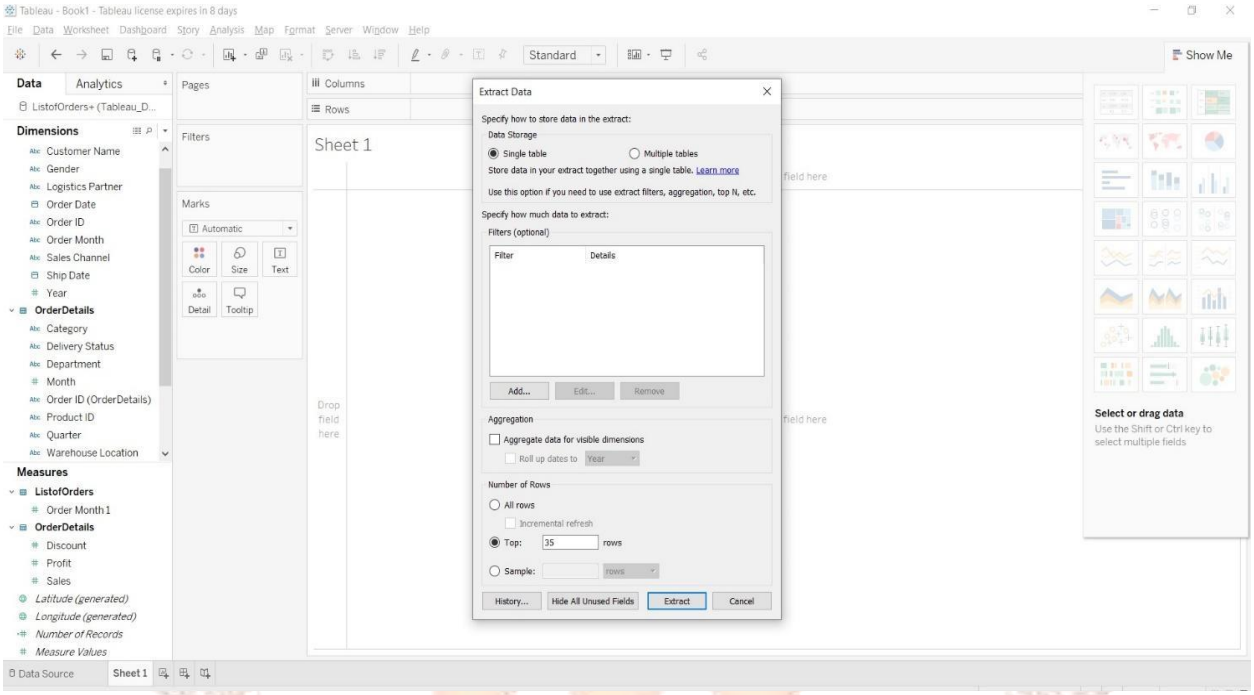


Fig 7.4: Extract 35 rows from dataset

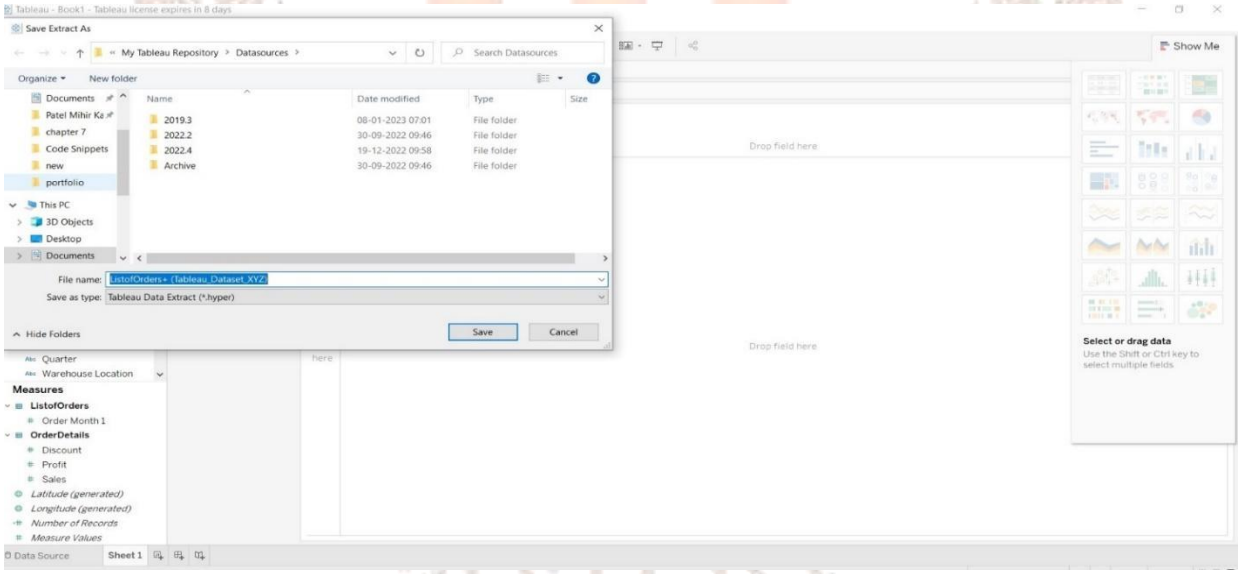


Fig 7.5: Save the Extracted Data file

For extracted data, the symbol is modified after extraction.

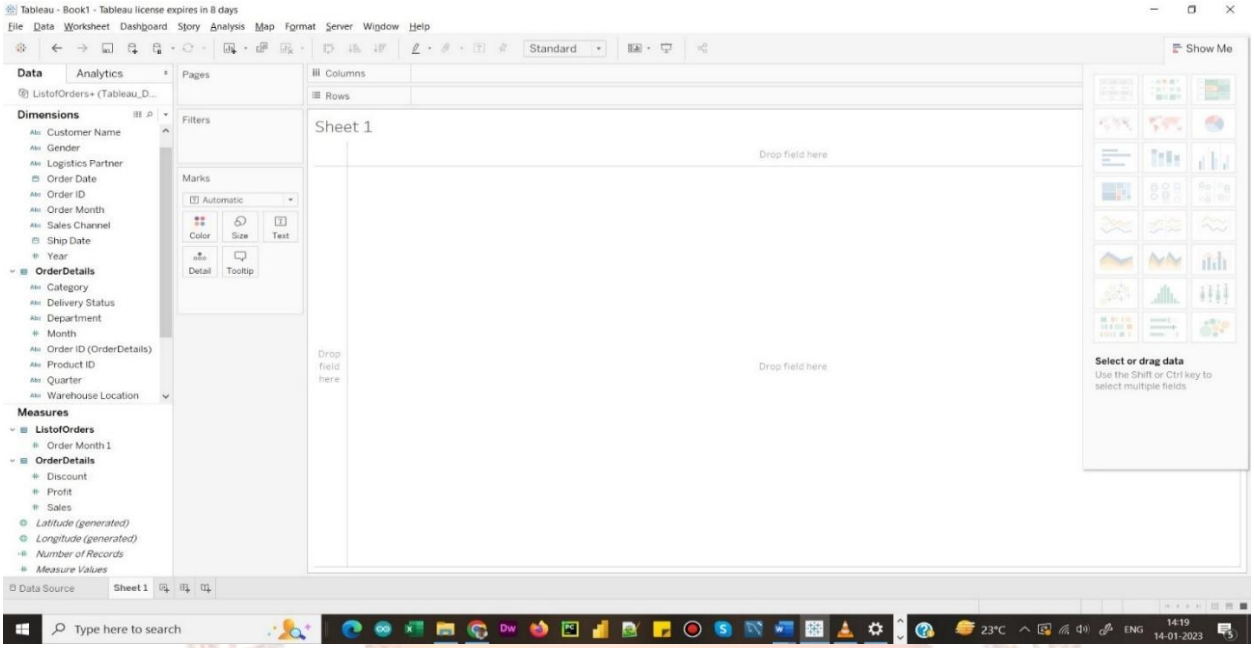


Fig 7.6: Extracted data with original data

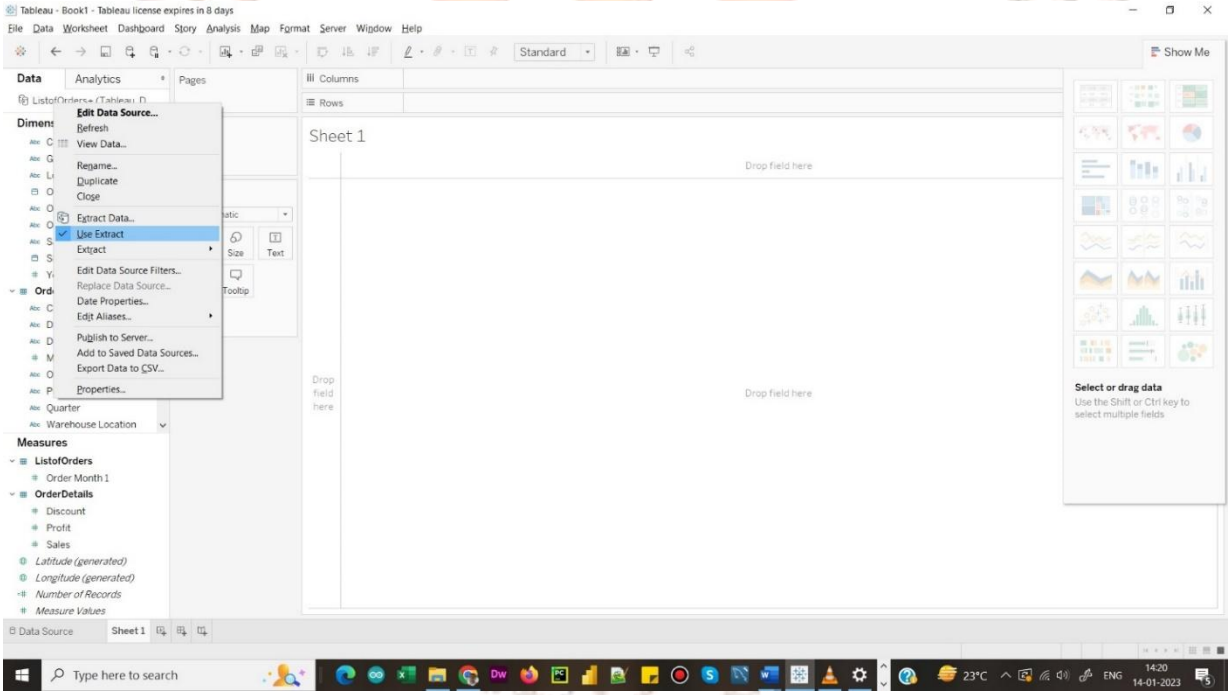


Fig 7.7: Use Extracted Data

You may now choose Extract then History to view the history of Extract.

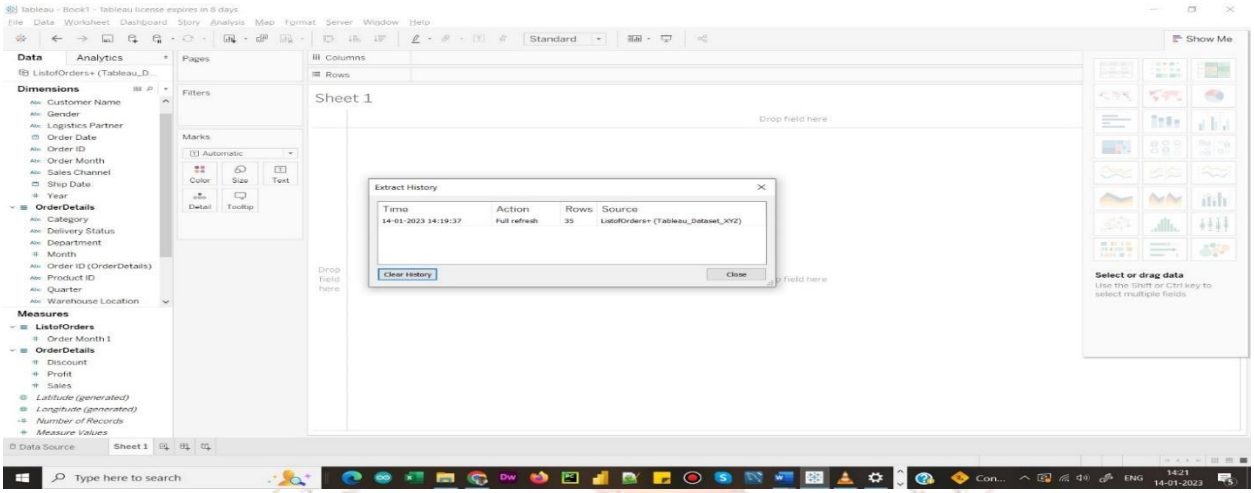


Fig 7.8: History of Extract

SELF-ASSESSMENT QUESTIONS – 1

1. A powerful feature of Tableau is _____.
2. _____is the Tableau File Extension.

3. REPLACE AND METADATA

Tableau captures the metadata details of the data source, such as the columns and data types, after connecting to it. This is what is used to generate the dimensions, measures, and calculated fields that are used in views. You can browse the metadata and modify some of its properties to meet your specific needs.

- Examining the Metadata:

Tableau displays all possible tables and columns in a data source after connecting to it.

- Changing the Data Type

If necessary, you can change the datatype of some of the fields. Tableau may fail to recognize the data type from the source depending on the nature of the source data. In such cases, we can manually change the data type.

- Renaming and concealment

The renaming option allows you to change the names of the columns. You can also hide a column so that it does not appear in your data view. By selecting the data type icon in the metadata grid, you can access these options.

- Column Alias

Each column in the data source can be given an alias, which aids in understanding the column's nature.

Practical steps with example:

Drag and drop fields from the data window into the worksheet's cards and shelves to create a data display.

Instead of manually creating views by dragging and dropping files, you can utilize show me to do so.

1) For better understanding, you might want to automatically build views.
2) To make time.

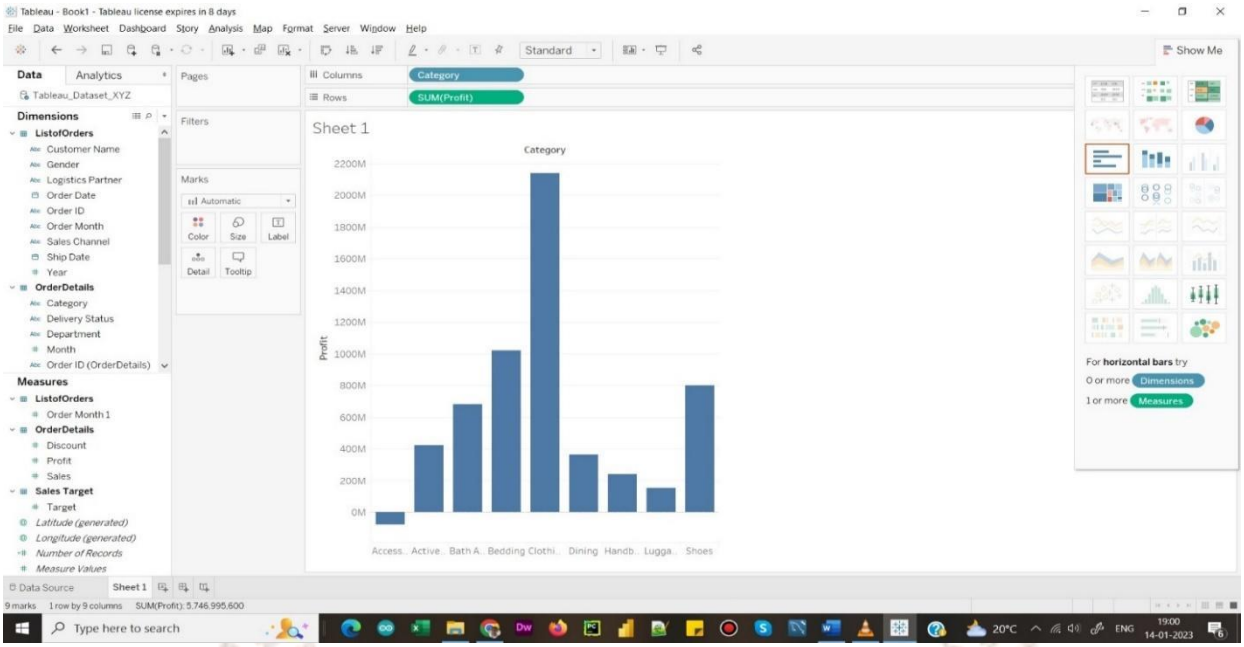


Fig 7.9: View of data

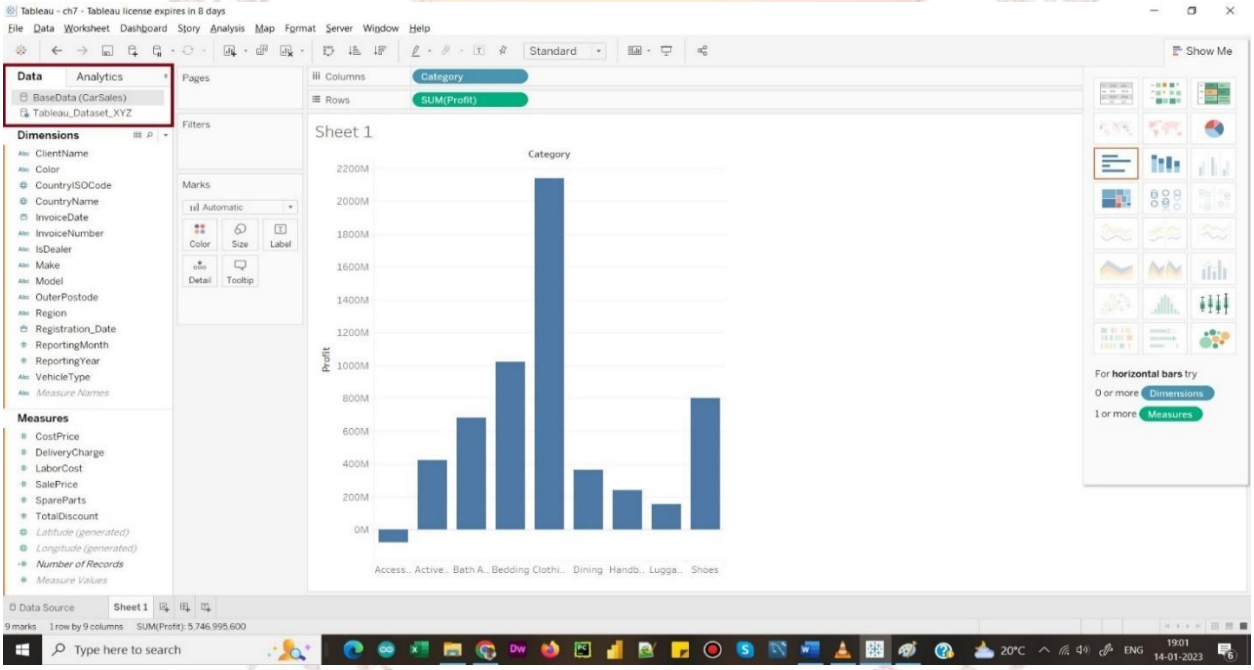


Fig 7.10: Data tab to replace Data Source

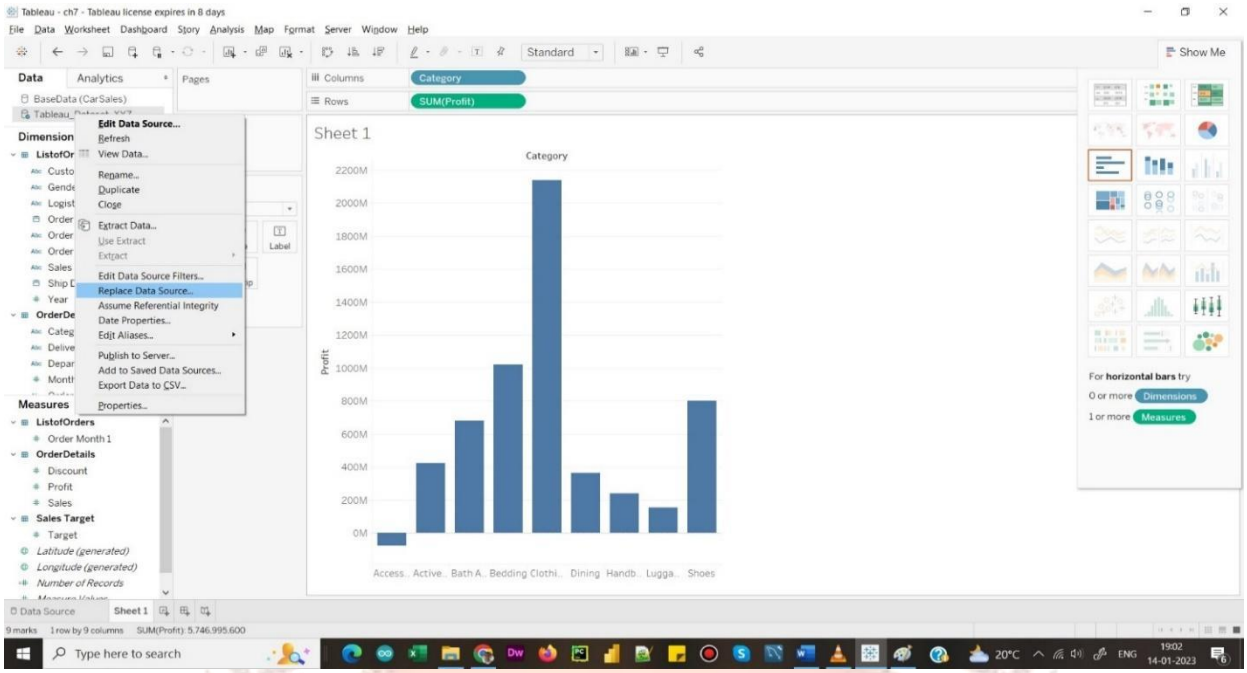


Fig 7.11: Replace Data Source

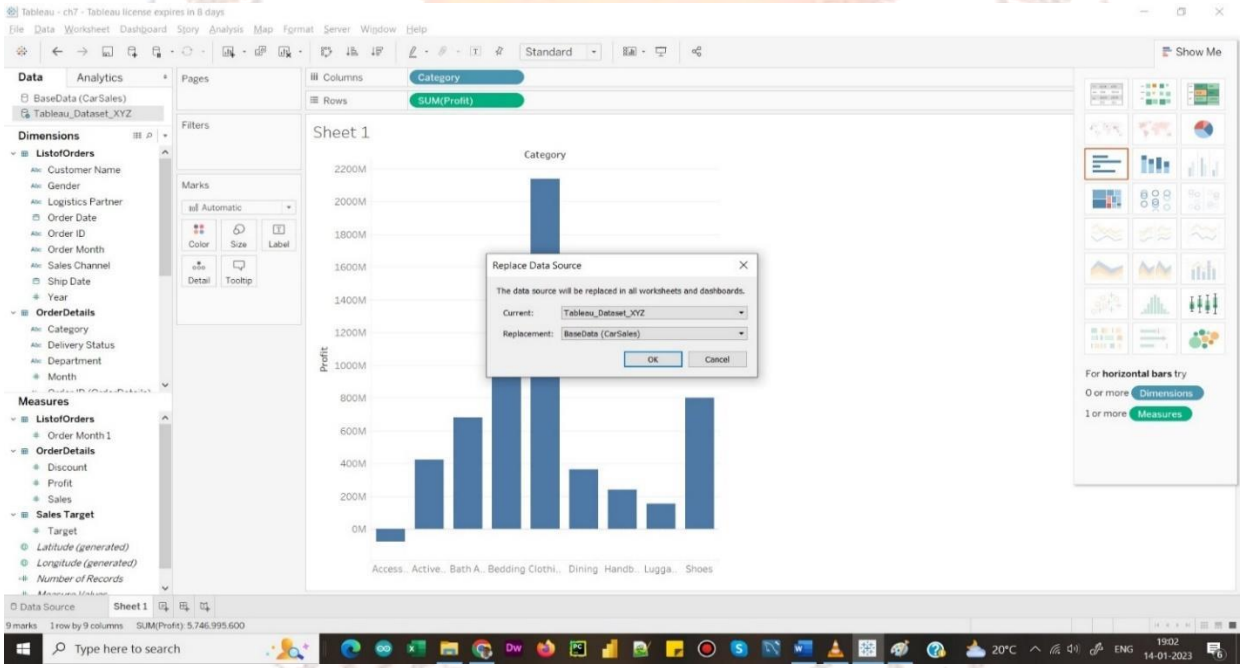


Fig 7.12: Select other Data Source

As you can see in the graphic, the rows and columns are not identifiable after changing the data source.

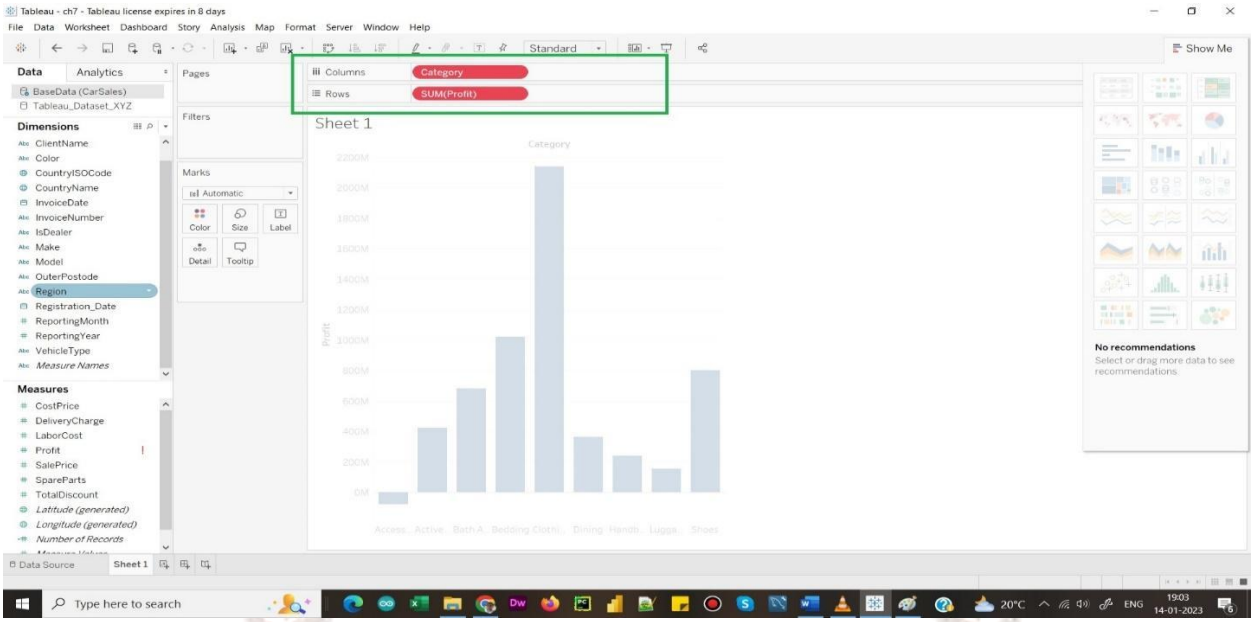


Fig 7.13: Rows and Columns not Identified

You can now alter the reference according to the requirement by going to the specific dimension or measure.

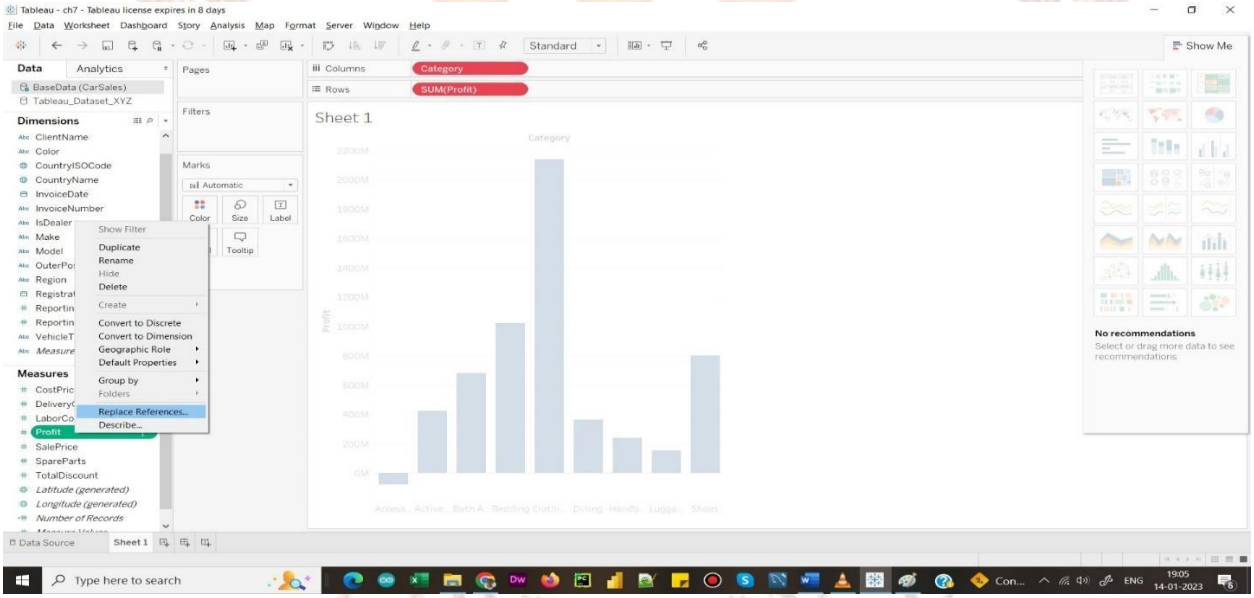


Fig 7.14: Change the references

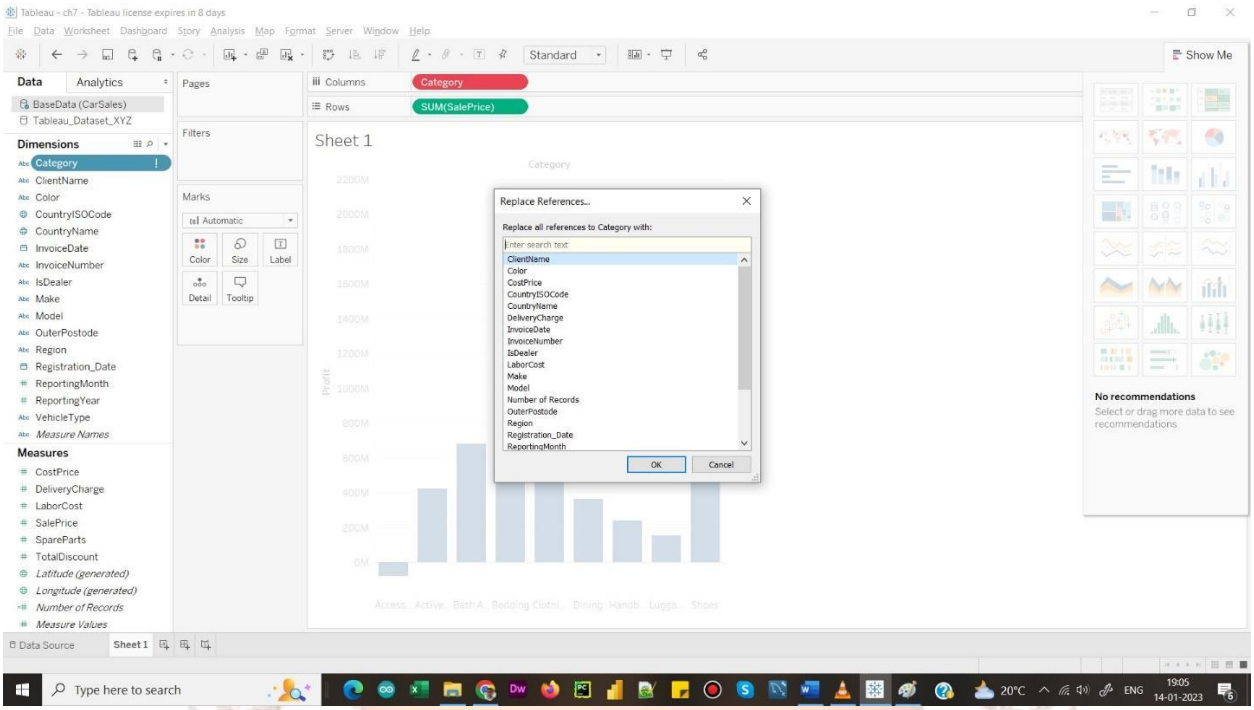


Fig 7.15: After Changing the references

SELF-ASSESSMENT QUESTIONS – 2

3. Can Data Blending be used to perform all kinds of joins? YES/NO
4. Is it possible to aggregate dimensions? YES/NO

4. BUILDING VIEWS AND FORMATTING VIEWS

A custom data view is used to supplement standard data views with additional features, allowing the view to provide different types of charts for the same underlying data. You can, for example, drill down a dimension field that is part of a pre-defined hierarchy to obtain additional values of the measures at a different granularity. The following are some of the most commonly used and important custom data views provided by Tableau.

Drill Down View

You usually need to know the result of analysis for the next or previous level of aggregation for dimension fields that are part of a hierarchy. For example, if you know the result for a quarter, you may be curious about the results for each month in that quarter, and you may even require the result for each week. This is an example of drilling down to a finer level of granularity from existing dimensions. Right-click a table header and select Drill Down from the context menu to drill down and up for individual dimension members in a hierarchy. Consider a bar chart where the dimension category is in the columns shelf and the measure Sales is in the rows shelf.

Swapping Dimensions

By swapping the positions of the dimensions, you can create a new view from an existing one. This has no effect on the values of the measures, but it does change their position. Consider a view for analyzing Profit for each year for each segment and product category. You can drag the vertical line at the end of the category column to the segment column by clicking and dragging it. The following screenshot depicts this action.

Data Joining

Data joining is a common requirement in all types of data analysis. You may need to join data from multiple sources or from different tables within the same source. Tableau includes the ability to join tables by using the data pane found under Edit Data Source in the Data menu.

Data blending

Tableau's Data Blending feature is extremely powerful. It is used when you want to analyze related data from multiple data sources in a single view. Consider the following scenario: Sales data is stored in a relational database, and Sales Target data is stored in an Excel

spreadsheet. To compare actual sales to target sales, you can now blend the data using common dimensions to access the Sales Target measure. Primary and secondary data sources are the two sources used in data blending. A left join is formed between the primary and secondary data sources, with all data rows from the primary and matching data rows from the secondary data source.

Practical steps are as follows:

After altering the reference, the names of the columns and rows are identified, and the graph is displayed.

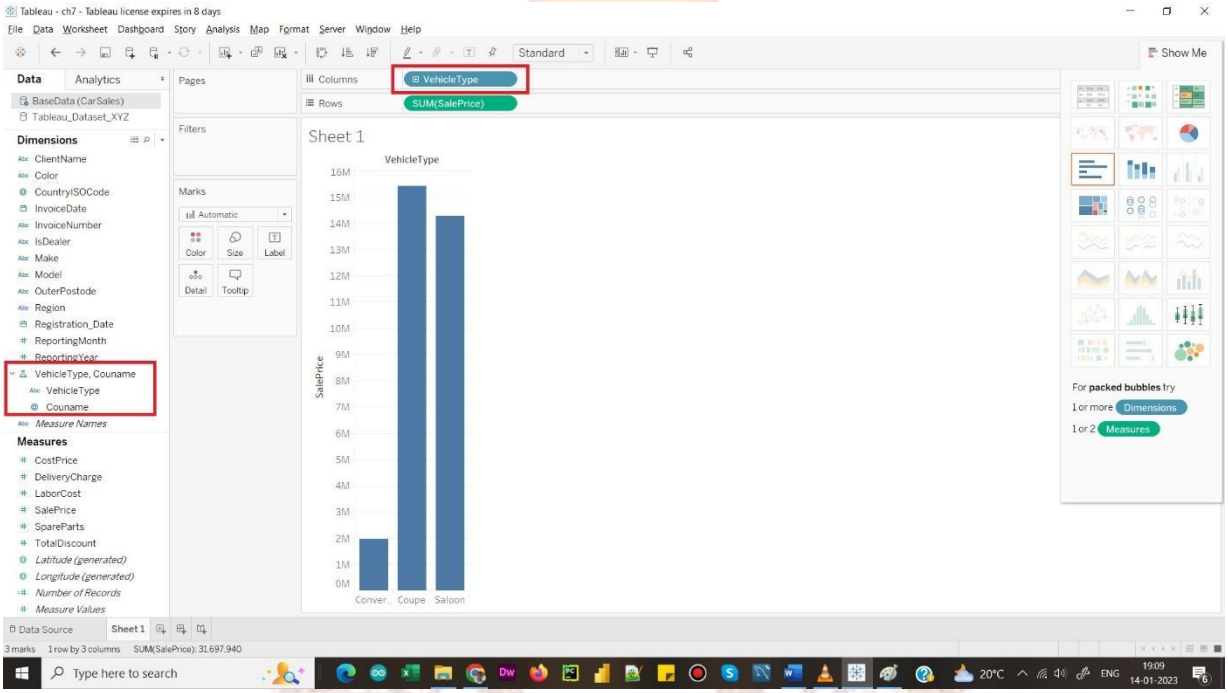


Fig 7.16: Graph enabled

You can take multiple column in the column based on visualization.

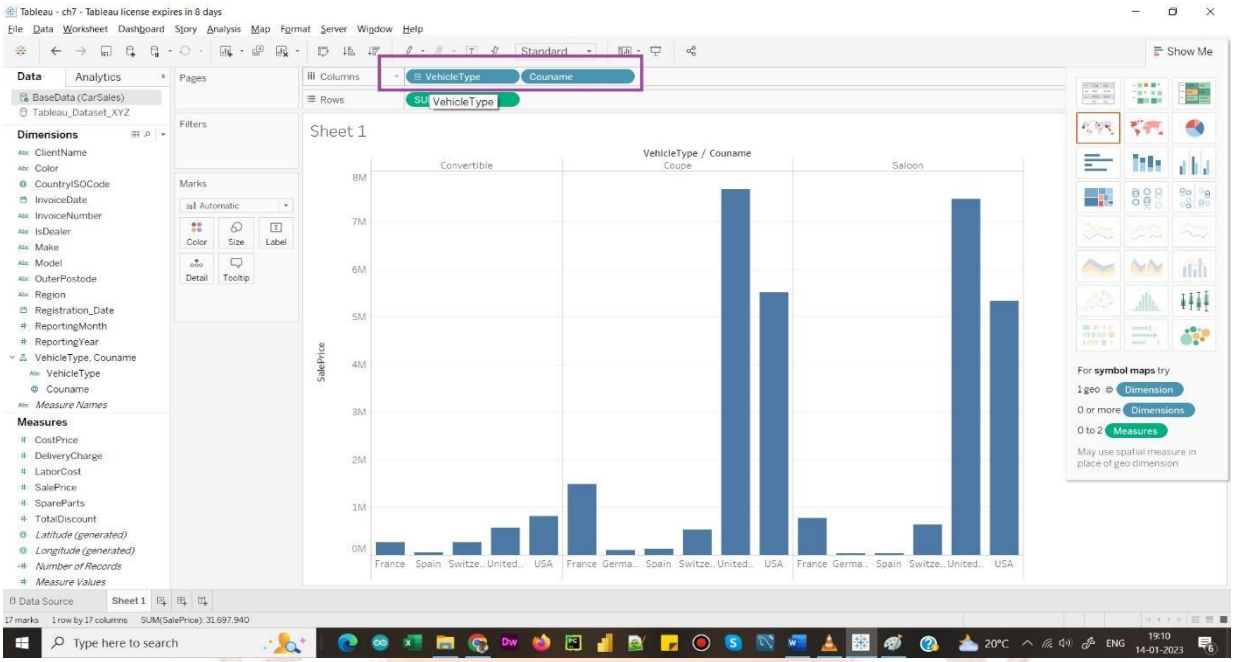


Fig 7.17: Multiple attributes on the axis

In addition to this, you can make a folder for a certain kind of work. For instance, you can make a folder for time-related information.

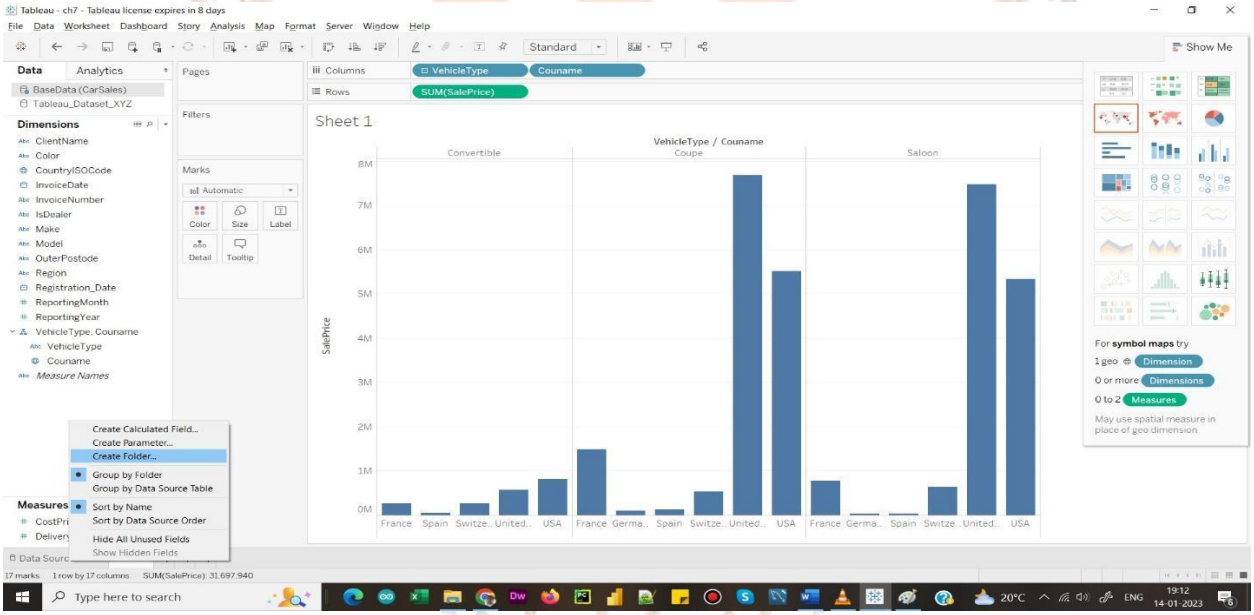


Fig 7.18: Create a Folder for Specific attributes

You may also change the type of data below.

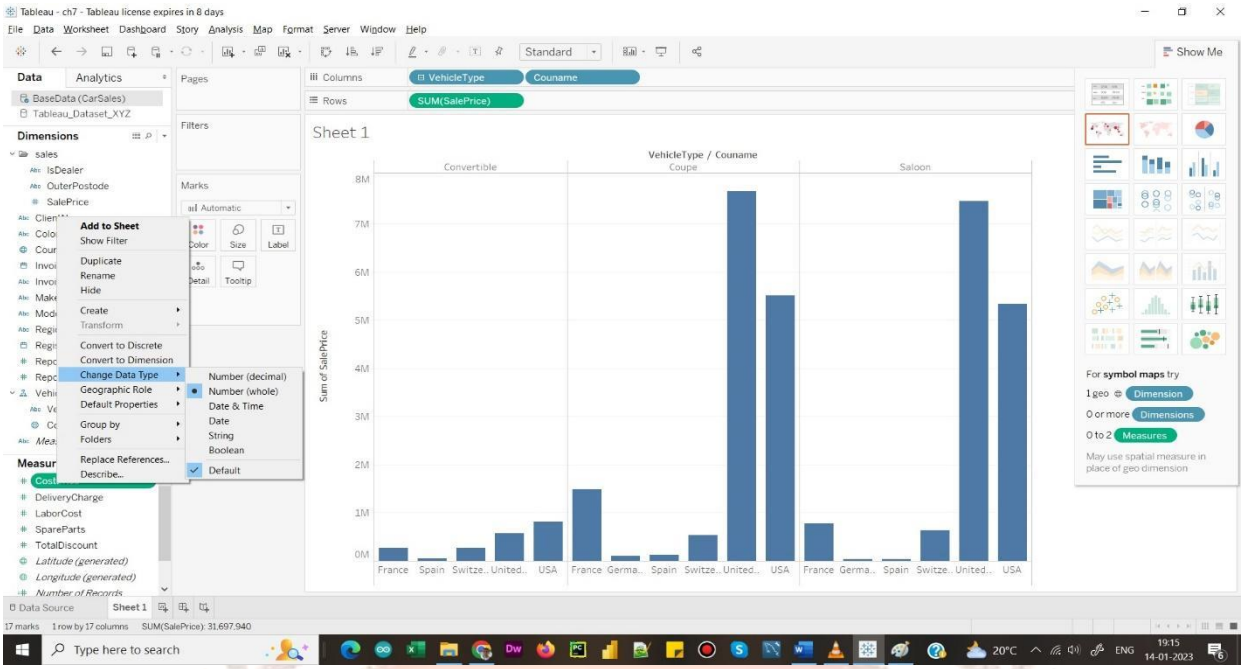


Fig 7.19: Change the type of data

Self-Assessment Questions - 3

5. What Type of Join Is Used in Blending?

You can now drag several measures and dimensions (attributes) to the axis in columns or rows.

When you select an attribute on an axis in Tableau, it will recommend the type of visuals that would fit the data the best from the show me alternatives.

whether it is a table, pie chart, bar graph, etc.

Here you can select color from the default properties to change the color of the visuals.

You can select aggregation type min, max, sum, count for the visuals.

You can add labels to drag the specific attributes to the text marks for the visuals.

6. Explain types of data sources supported in tableau.

Live Connection

The Connect Live feature is used to analyse data in real time. Tableau connects to a real-time data source and continues to read the data in this case. As a result, the analysis result is up to the second, and the most recent changes are reflected in the result. However, the source system is burdened because it must continue to send data to Tableau.

In-Memory

Tableau can also process data in-memory by caching it in memory and not connecting to the source while analysing it. Of course, the amount of data cached will be limited by the amount of memory available.

Combine Data Sources

Tableau can connect to multiple data sources at once. For example, you can define multiple connections in a single workbook to connect to a flat file and a relational source. This is used in data blending, which is a very unique Tableau feature.

7. Explain types of metadata operations supported in tableau?

Tableau captures the metadata details of the data source, such as the columns and data types, after connecting to it. This is what is used to generate the dimensions, measures, and calculated fields that are used in views. You can browse the metadata and modify some of its properties to meet your specific needs.

- Examining the Metadata:

Tableau displays all possible tables and columns in a data source after connecting to it.

- Changing the Data Type

If necessary, you can change the datatype of some of the fields. Tableau may fail to recognize the data type from the source depending on the nature of the source data. In such cases, we can manually change the data type.

- Renaming and concealment

The renaming option allows you to change the names of the columns. You can also hide a column so that it does not appear in your data view. By selecting the data type icon in the metadata grid, you can access these options.

- Column Alias

Each column in the data source can be given an alias, which aids in understanding the column's nature.

8. Explain view in tableau and which type of views are supported in tableau

A custom data view is used to supplement standard data views with additional features, allowing the view to provide different types of charts for the same underlying data. You can,

for example, drill down a dimension field that is part of a pre-defined hierarchy to obtain additional values of the measures at a different granularity. The following are some of the most commonly used and important custom data views provided by Tableau.

1. Drill Down View

You usually need to know the result of analysis for the next or previous level of aggregation for dimension fields that are part of a hierarchy. For example, if you know the result for a quarter, you may be curious about the results for each month in that quarter, and you may even require the result for each week. This is an example of drilling down to a finer level of granularity from existing dimensions. Right-click a table header and select Drill Down from the context menu to drill down and up for individual dimension members in a hierarchy. Consider a bar chart where the dimension category is in the columns shelf and the measure Sales is in the rows shelf.

2. Swapping Dimensions

By swapping the positions of the dimensions, you can create a new view from an existing one. This has no effect on the values of the measures, but it does change their position. Consider a view for analyzing Profit for each year for each segment and product category. You can drag the vertical line at the end of the category column to the segment column by clicking and dragging it. The following screenshot depicts this action.

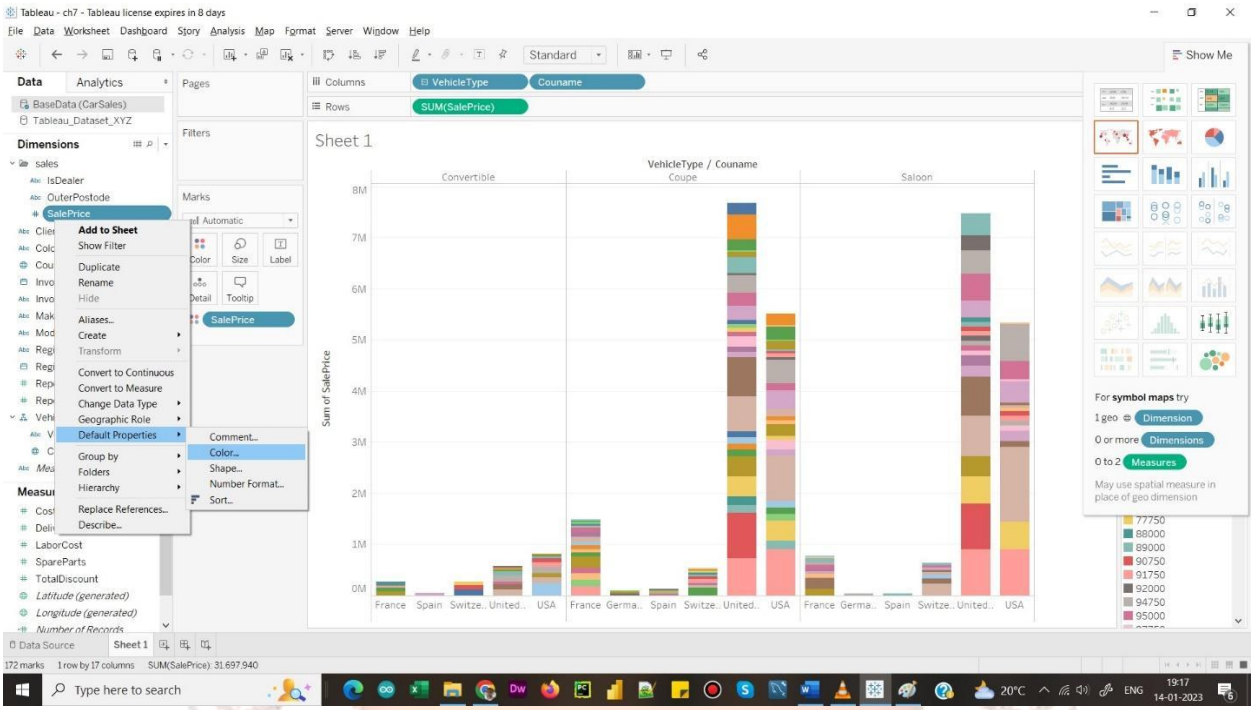


Fig 7.20: Change color of visuals

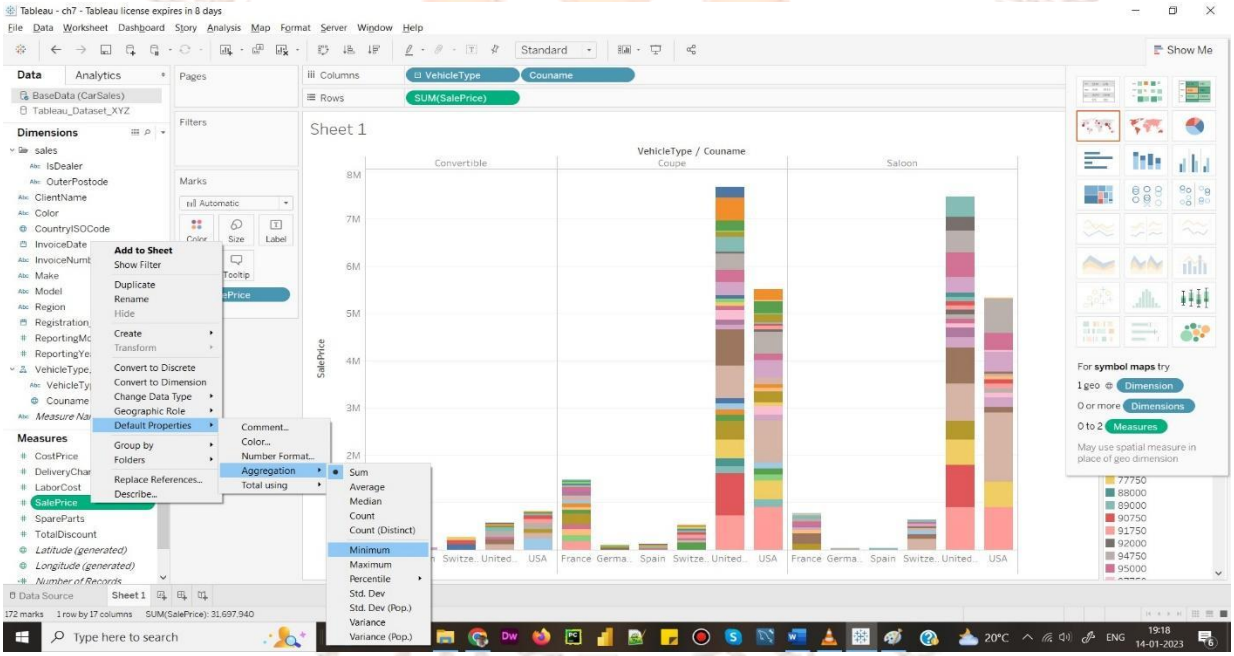


Fig 7.21: Change Aggregation type

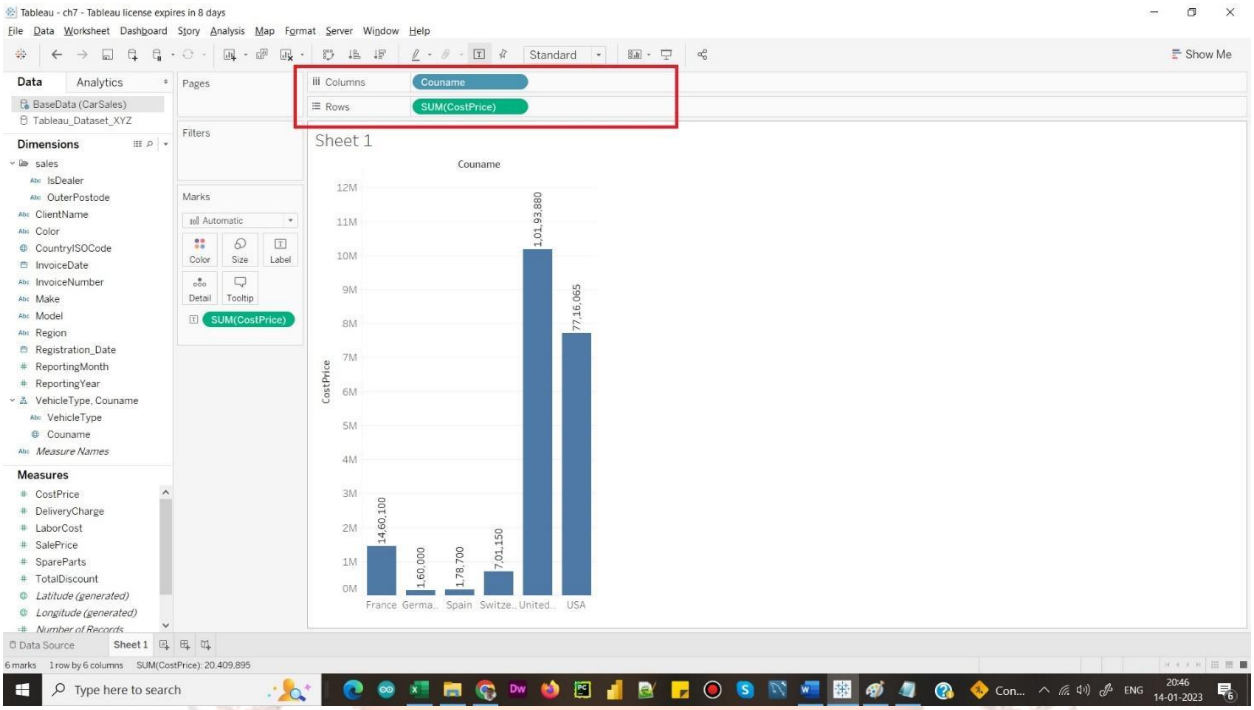


Fig 7.22: Changes saved to Data Sources

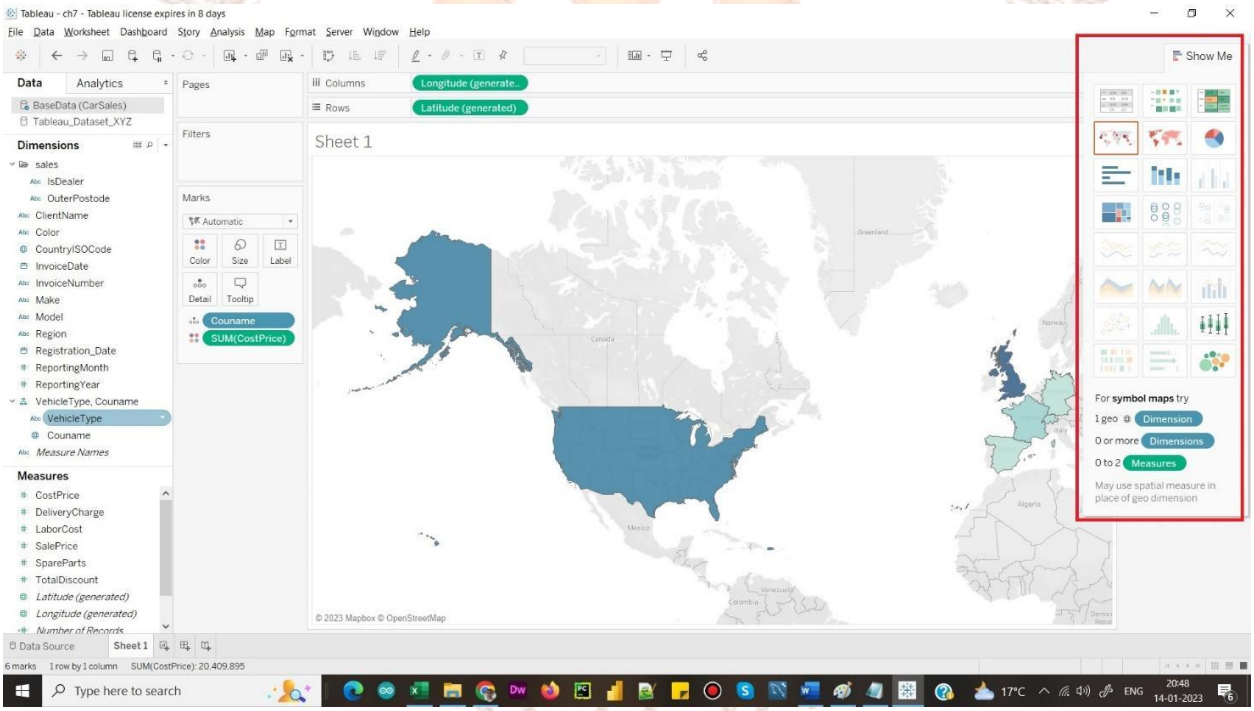


Fig 7.23: Add label to visuals

You can View the map for different regions here show me option to change the visual type.

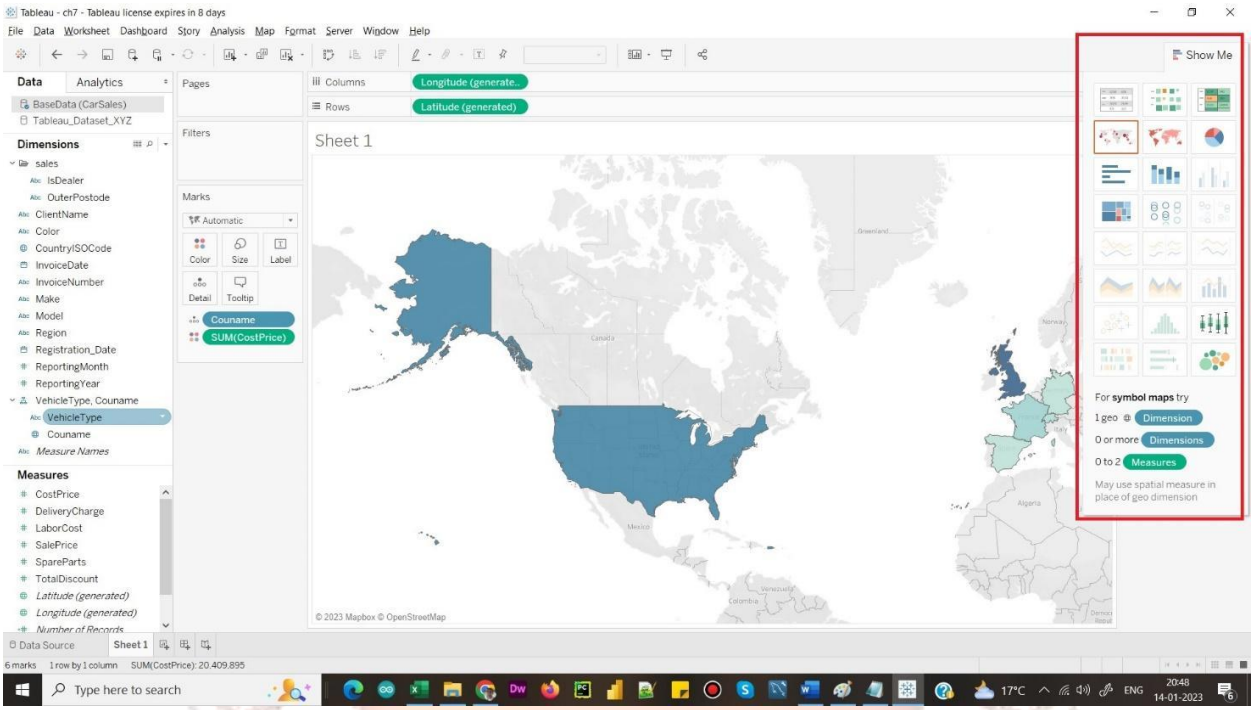


Fig 7.24: Map Visual for Data

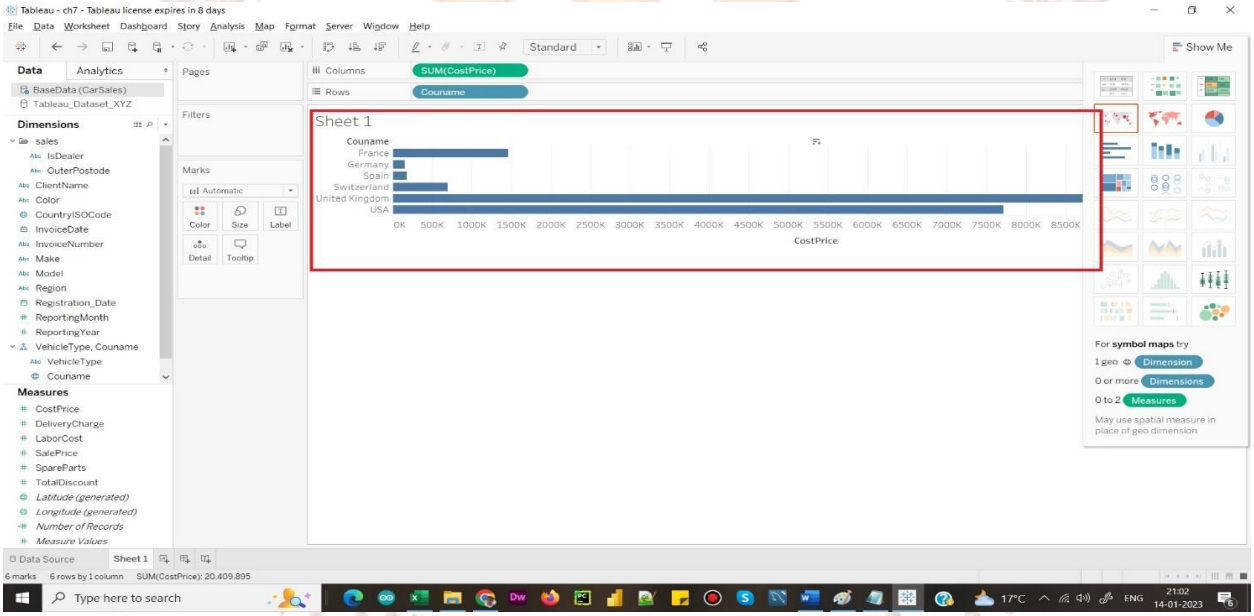


Fig 7.25: Horizontal bar Visual for Data

By clicking on the label and selecting "Show Mark Label" you may alter the label's size and color as desired.

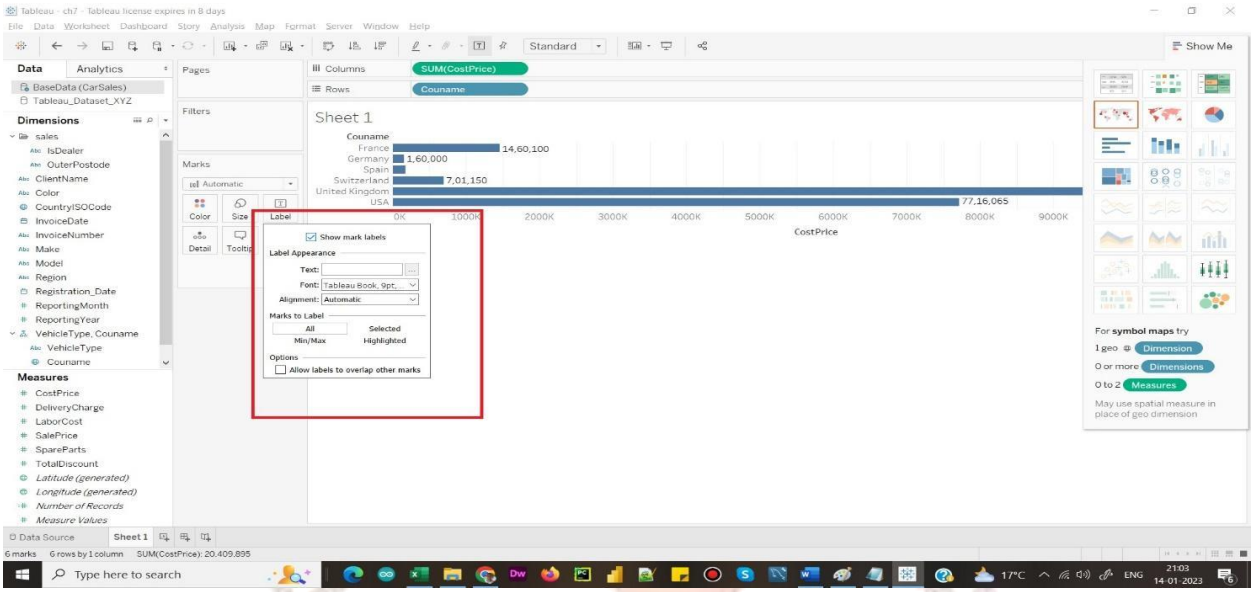


Fig 7.26: Horizontal bar Visual for Data

By clicking on the color and selecting "font and shading" you may alter the font's type and color as desired.

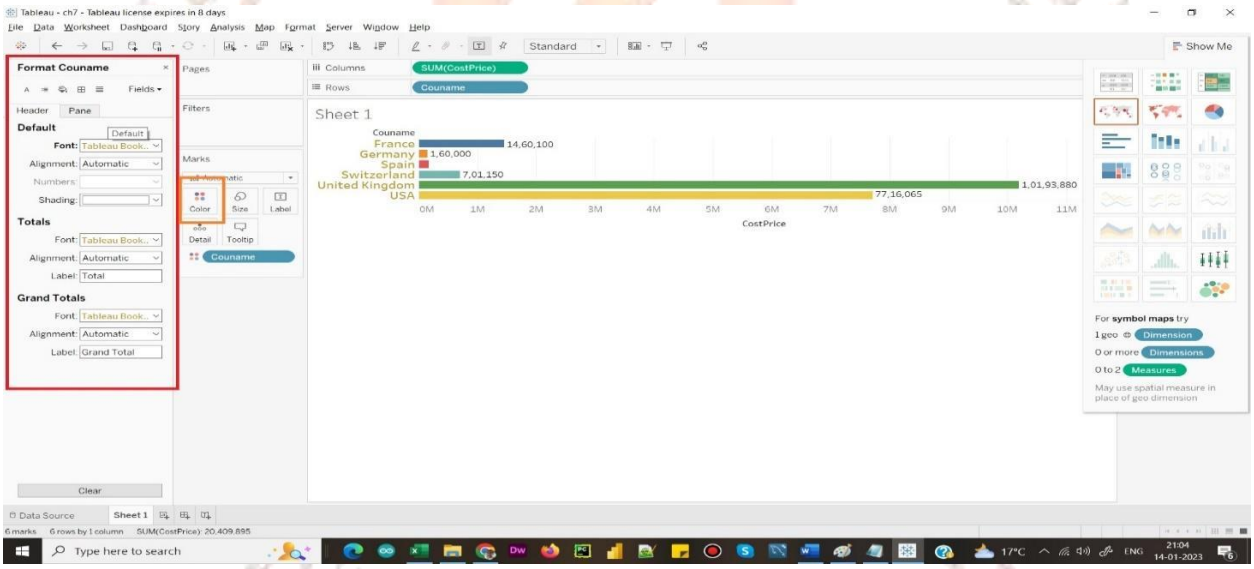


Fig 7.27: Changing color of visuals

By clicking on the filter and selecting TOP you may extract top 3 or 5 data for visuals as per condition.

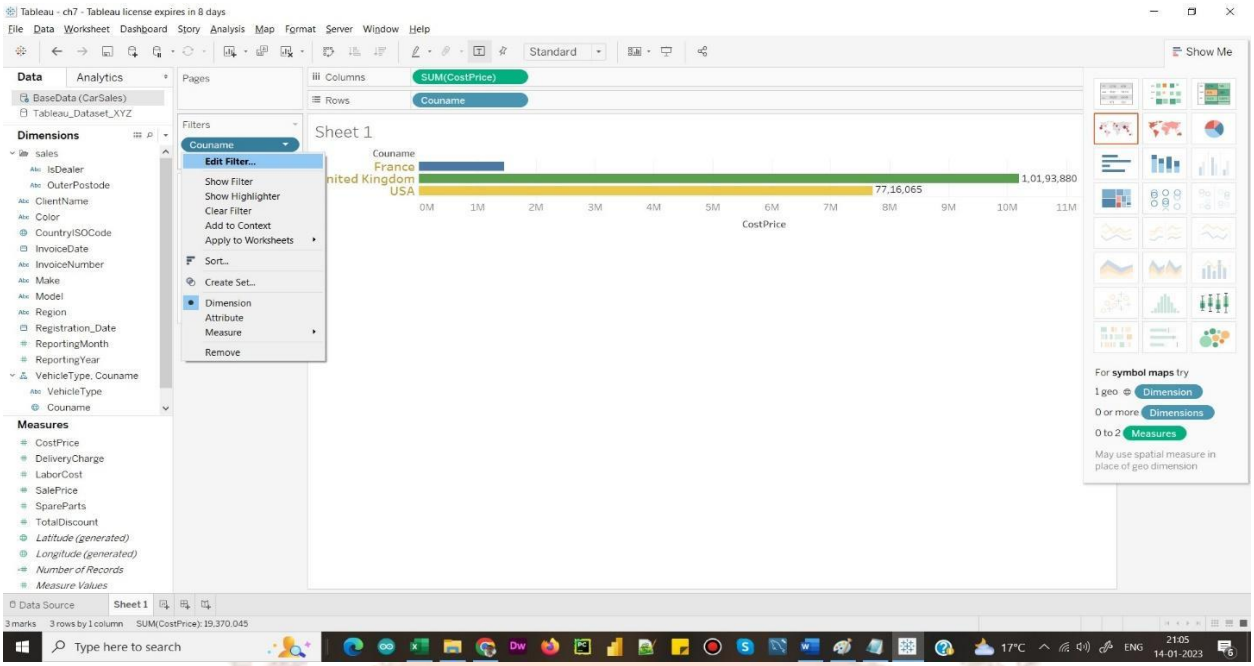


Fig 7.28: Filter Selection

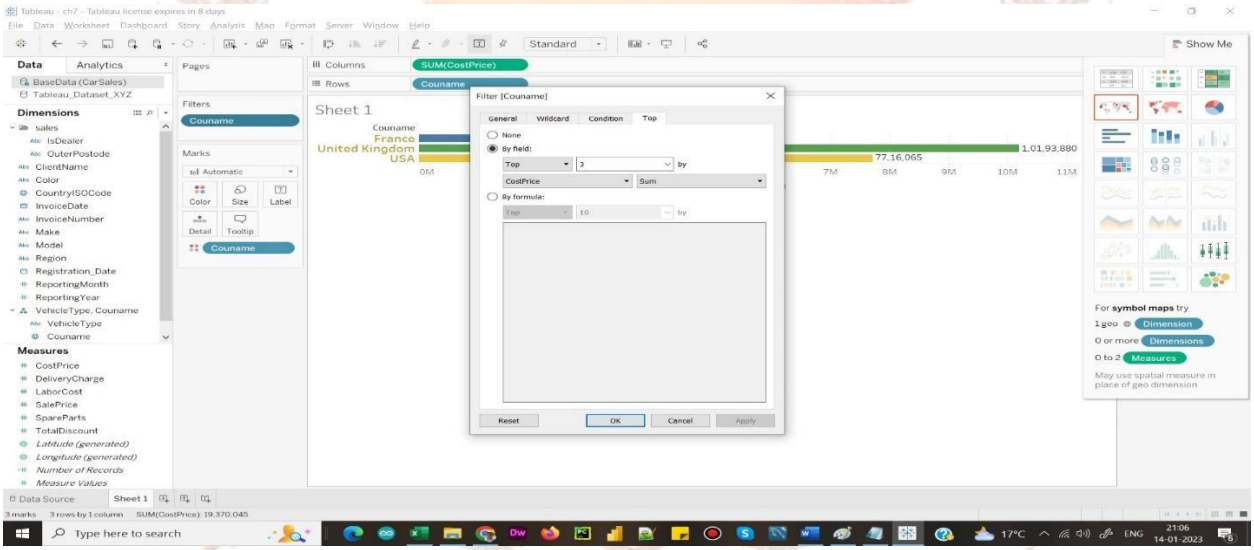


Fig 7.29: Filter Selection for TOP records

5. CASE STUDIES

J.P. Morgan Chase & Co. is a leading multinational bank and financial services company based in the USA. It is the largest investment bank in the US and sixth largest in the world.

The challenge

The amount of data produced by the company increased with increasing growth and expansion of business of the firm because of successful mergers and acquisitions. This led to the need for a robust self-service data governance and analytics solution. Initially, JPMC had an IT-owned analytics set-up which they planned to change into a business-owned system12. The IT department used standard tools like Excel and SQL Server for data analytics and reporting. But these tools were eventually proved inefficient as they caused confusion and obscurity in data governance due to data replication.

After this, the company switched to BI tools Cognos and Business Objects, but those tools too were not able to meet the company's requirement. What the company needed was a data governance tool taking care of business aspects like data access, data analysis, IT governance, and business priorities of the team.

Implementation

JPMC deployed Tableau as its core self-service data governance tool at an enterprise level. JPMC with the help of COE (The Center of Excellence) team facilitated the process of Tableau adoption to their users by recruiting a team of 8 trainers who trained 1200 new developers and analysts to work on Tableau. The training program involved online as well as classroom learning sessions. The initial user base was 400 Tableau Server users in 2011 which has grown into a family of 30,000 users today.

Skilled Tableau business users work in teams into different departments and sectors of JPMC. Currently, there are about 500 teams using Tableau for analytical and data governance purposes.

The changes

Deploying Tableau for data governance in JP Morgan Chase was a successful step as it brought many positive changes.

- The marketing operations team could analyze customer data to track customer journey and preferences which helped them in deciding website design, promotional materials and launching new products like Chase mobile app.
- Financial and branch managers used Tableau apps to analyze customer data in order to provide better customer banking experience.
- Tableau has given self-service analytical capabilities to a whole lot of people taking care of different operations such as traders, risk analysts, compliance team, operations analysts, sales analysts, etc.
- JPMC was able to reduce manual reporting time. Earlier, the team took months to create reports but with the help of Tableau, detailed reports were made in weeks. Tableau has saved a lot of the company's valuable time and has shifted the focus from report generation to analyzing the reports, gaining meaningful insights into data and efficient decision-making.
- Tableau has enabled JPMC to establish stronger customer relationships by integrating customer's data with the line of business aspects such as products, marketing, services and creating common data sets. Thus, Tableau acts as a front-end tool to maintain customer relations.
- The marketing teams at JPMC can analyze population data using Tableau to determine optimal targets to launch new campaigns.
- To ensure customer satisfaction by analyzing customer activities and behavior using call center metrics and website analytics.
- JPMC's retail branches also use Tableau dashboards to gain a better understanding of the market and improve their business.
- Tableau successfully created a bridge between IT and business by providing apps and dashboards for risk analysis and compliance data usage. It also functions as per the government regulations.

6. TERMINAL QUESTIONS

1. How to extract Data in Tableau?

Tableau creates a subset of data from the data source using data extraction. By implementing filters, this is helpful in boosting performance. Additionally, it aids in adding Tableau features to data that may not be present in the data source, such as identifying different values in the data. The data extract tool is typically used, though, to produce an extract that will be saved to the local drive for offline access by Tableau.

By selecting Data > Extract Data from the menu, data can be extracted. It offers a variety of options, including imposing constraints on the number of rows to be extracted and choosing whether to aggregate data for dimensions. The Extract Data option is displayed on the following screen.

You can design filters that will only return the pertinent rows if you want to extract a subset of data from the data source. Create an extract using the Sample Superstore data set as our example. Select the list under the filter option, then check the box next to the value you want to pull from the source when the data is pulled.

2. How to edit metadata in Tableau?

Tableau records the metadata information of the data source, including the columns and their data types, after connecting to the data source. The dimensions, measurements, and calculated fields used in views are produced using this. You can look through the metadata and modify some of its characteristics according to your needs.

When Tableau connects to a data source, it displays all the tables and columns that could be included in the source. To check the metadata, think about the source "Sample Coffee shop." Select "Connect to a data source" from the Data menu. Search for the "Sample - Coffee shop" MS Access file. Drag the Product table onto the data canvas. The following screen, which displays the column names and their data types, appears after selecting the file.

3. Explain Building views and formatting views in tableau.
4.

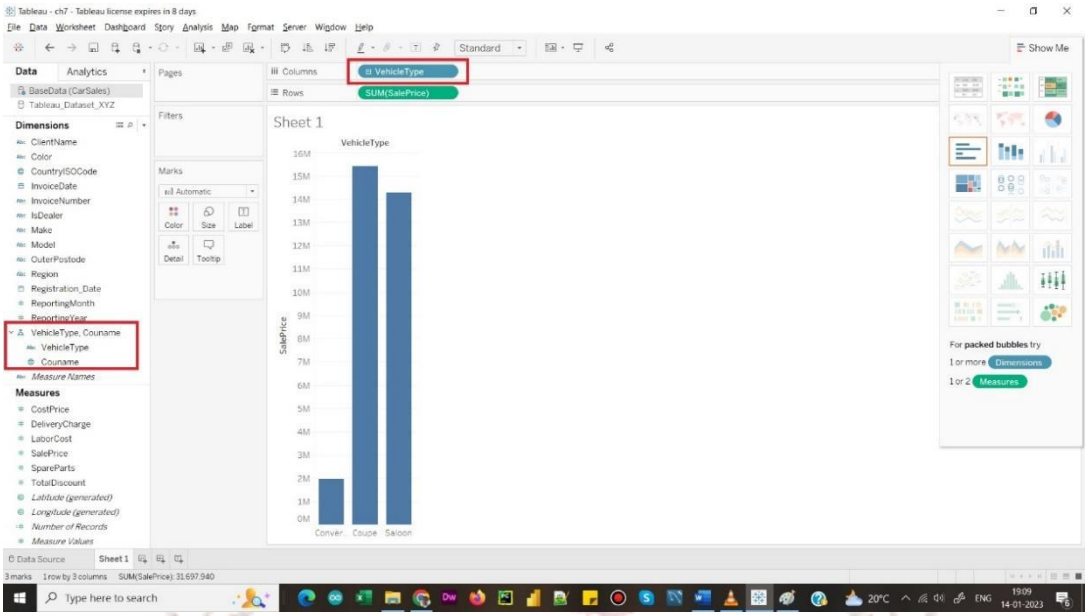
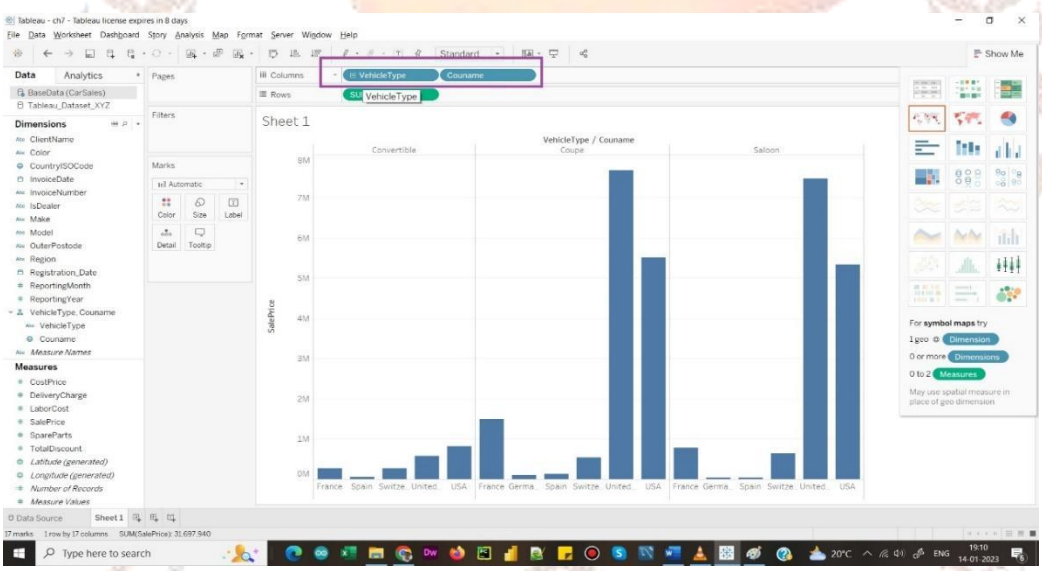


Fig 7.30: Graph enabled

5. You can take multiple column in the column based on visualization.



6.

Fig 7.31: Multiple attributes on the axis

7. In addition to this, you can make a folder for a certain kind of work. For instance, you can make a folder for time-related information.

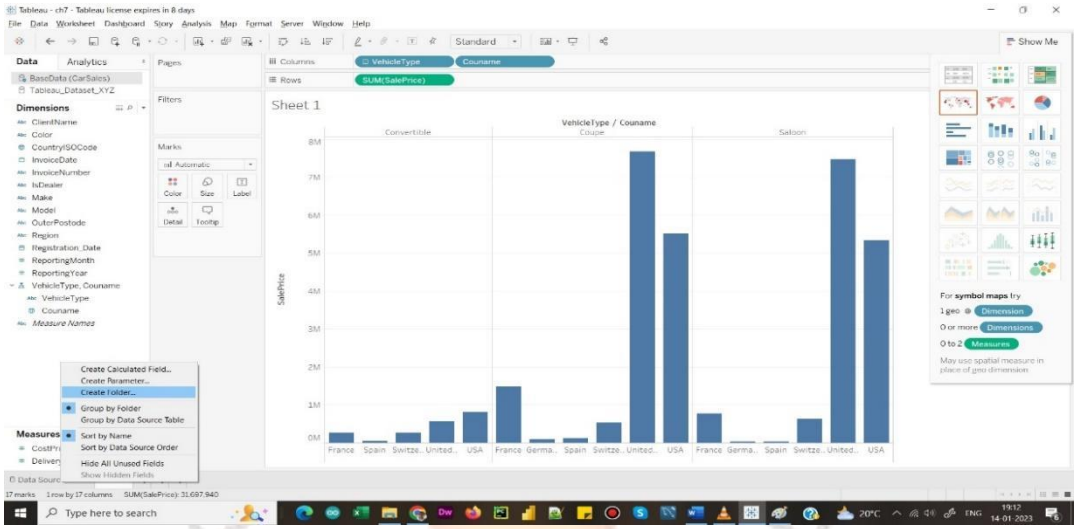
8. 

Fig 7.32: Create a Folder for Specific attributes

9. You may also change the type of data below.

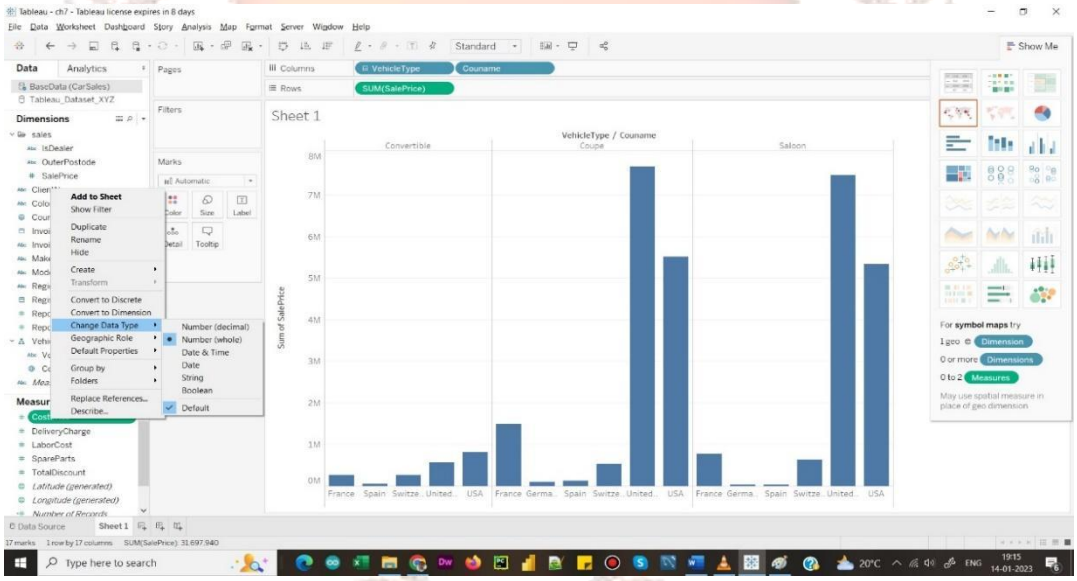
10. 

Fig 7.33 Change the type of data

7. ANSWERS

Self-Assessment Questions

1. Data blending
2. twbx
3. YES
4. No
5. Left join

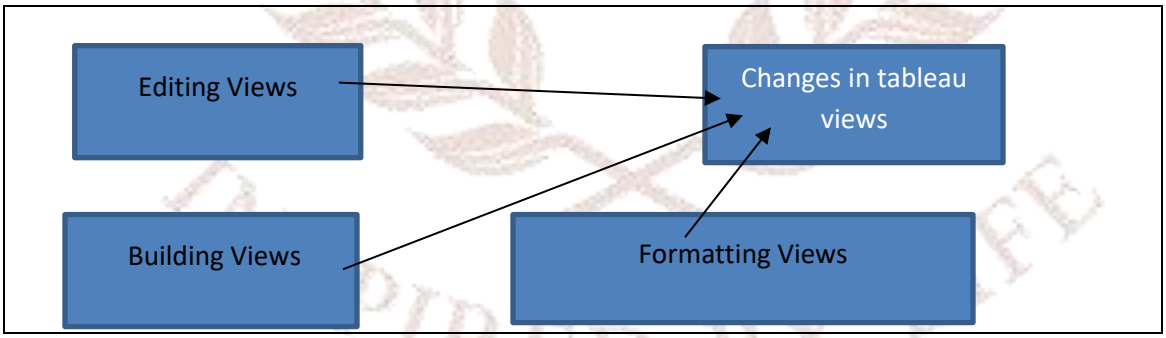
8. SUMMARY

All the most well-liked data sources are accessible through Tableau. The data sources listed below can be connected to using Tableau's native connectors. file types including Excel, CSV, and others. databases that are relational, including Oracle, SQL Server, DB2, and others. Platforms for cloud computing include Windows Azure, Google Big Query, and others.

9. GLOSSARY

- Visualization: finding insight from data
- Import data: Load the data into tableau for processing
- Replace Data: Change the data source

10. CONCEPT MAP



11. REFERENCE

- Practical Tableau: 100 Tips, Tutorials, and Strategies from a Tableau Zen Master
- Tableau Your Data! Fast and Easy Visual Analysis with Tableau Software

INSPIRED BY LIFE